Learning Multi-Purpose Speech Representations via Self-Supervised Learning — WavLM

Chen et al. (2022), in their paper *WavLM: Large-Scale Self-Supervised Pre-Training for Full Stack Speech Processing*, present a framework for learning speech representations by means of Self-Supervised Learning (SSL), intended for a wide range of speech processing tasks and not only for recognizing linguistic content. Unlike previous approaches that focused mainly on Automatic Speech Recognition (ASR), the study emphasizes the need to learn representations that contain information about the content of the speech, the speaker's identity, and additional acoustic and paralinguistic features. The researchers point out that different tasks require different types of information, and sometimes even pose conflicting requirements: speech recognition focuses on the linguistic content, while speaker verification requires preserving unique features of the speaker's identity.

The architecture of WavLM is based on a Convolutional Feature Encoder and a Transformer, and mainly extends HuBERT's Masked Speech Prediction framework. The central novelty is combining Masked Speech Prediction with Denoising at the Pre-training stage. Within this framework, the researchers create artificially corrupted recordings by means of adding background noise or overlapping speech of an additional speaker, and train the model to predict the Pseudo-labels of the original speech at masked segments. The goal of the process is to improve the model's ability to learn useful representations even when the speech signal contains interference or several speakers. In addition, the researchers combine a Gated Relative Position Bias mechanism, which makes it possible to adapt the effect of relative position in the Self-Attention mechanism to the content of the representations, and expand the training dataset to about 94,000 hours of audio from diverse sources, including Libri-Light, GigaSpeech and VoxPopuli.

The model was evaluated by means of the SUPERB framework, which includes 15 tasks representing different aspects of speech processing, and also by means of four additional tasks that were examined on dedicated evaluation datasets. In the SUPERB experiments, the WavLM Base model, which includes about 94.7 million parameters and was trained on 960 hours of audio, achieved better results than wav2vec 2.0 Base and HuBERT Base on several central tasks. For example, on the Speaker Diarization Error Rate metric a value of 4.55% was obtained, compared with 6.08% and 5.88%, respectively. In an Ablation experiment, removing the Denoising component increased the Diarization error to 6.03%, while removing the architectural change increased the WER on the ASR task from 6.21% to 6.80%. These results indicate a different contribution of the model's components to different tasks, and not a uniform improvement in every metric.

In the additional evaluation experiments, WavLM Large showed strong performance in speaker verification, speech separation and Speaker Diarization. Among other things, a speaker verification system based on WavLM Large, after further adaptations of Fine-Tuning and score calibration, achieved an EER of 0.383% on the VoxCeleb1-O protocol. It is important to emphasize that this result refers to speaker identity verification, and not to voice forgery detection. In addition, the researchers showed that on some of the speaker and speech-separation tasks, using a frozen Encoder yielded better performance than updating its weights, a finding indicating that Fine-Tuning does not guarantee an improvement under all conditions. An analysis of the contribution of the Transformer layers also showed that different tasks benefit from different layers: early and middle layers contributed significantly to speaker-related information, while higher layers were more significant for content and speech-recognition tasks.

Despite its achievements, the study does not directly examine Deepfake Detection, and therefore does not prove that WavLM preserves the acoustic artifacts required to distinguish between real and fake speech. Moreover, the pre-training for robustness to noise and overlapping speech may help generalization to different recording conditions, but it may also reduce sensitivity to subtle acoustic features relevant to forgery detection. In addition, the improvement in performance following the expansion of the Pre-training data does not make it possible to fully isolate the contribution of the diversity of audio sources, since the scope of training and additional factors also change.

For the present study, WavLM offers a relevant approach for examining pre-trained speech representations within the voice forgery detection task. Following wav2vec 2.0, the two models can be compared as Frozen Encoders under a uniform protocol, and it can be examined whether WavLM's multi-purpose representations improve classification performance on the EER metric. In addition, the findings of the layer analysis justify examining a combination of representations from several layers versus using only the last layer, as well as a comparison between a Frozen Encoder and Fine-Tuning. These experiments may help determine the degree of suitability of SSL representations for detecting TTS, Voice Conversion and Neural-Audio-Codec-based forgeries, while examining the generalization ability between the project's data sources.

Bibliographic source:

Chen, S., Wang, C., Chen, Z., et al. (2022). *WavLM: Large-Scale Self-Supervised Pre-Training for Full Stack Speech Processing*. arXiv:2110.13900. https://arxiv.org/abs/2110.13900